Networks

Paths

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Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

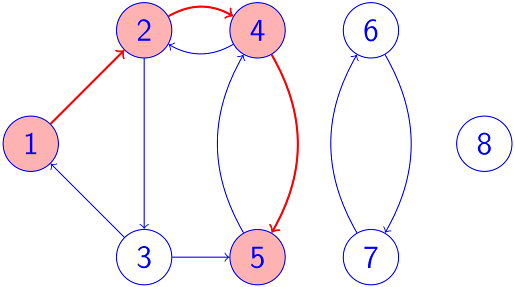
Paths

Networks are designed to connect elements.
The concept of paths describe how two elements in the network can be connected to each other.

Paths

Simple forward path

$$1 \to 2 \to 4 \to 5$$
$$P^{\to} = (1,2),(2,4),(4,5),$$
$$P^{\leftarrow} = \emptyset$$

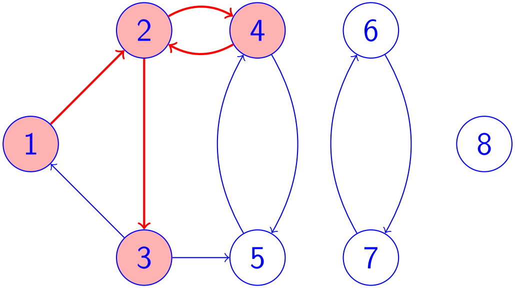


Forward path

$$1 \to 2 \to 4 \to 2 \to 3$$

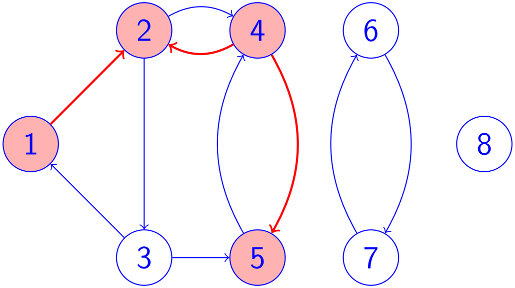
$$P^{\to} = (1,2), (2,4), (4,2), (2,3),$$

$$P^{\leftarrow} = \emptyset$$



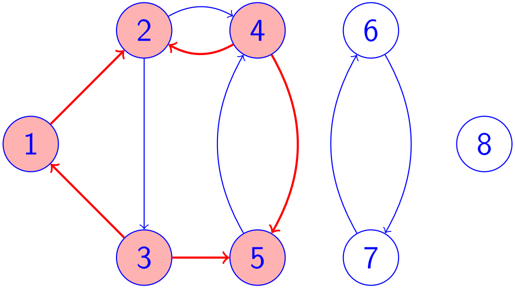
Simple path

$$1 \to 2 \leftarrow 4 \to 5$$
$$P^{\to} = (1,2),(4,5)$$
$$P^{\leftarrow} = (4,2)$$



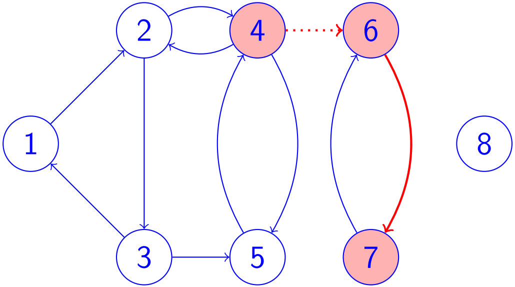
Simple cycle

$$1 \rightarrow 2 \leftarrow 4 \rightarrow 5 \leftarrow 3 \rightarrow 1$$
$$P^{\rightarrow} = (1,2),(4,5),(3,1)$$
$$P^{\leftarrow} = (4,2),(3,5)$$



Invalid path

$4 \to 6 \to 7$



Summary

Paths are connecting two nodes in a network: an origin and a destination
They are simple if no node is repeated
They are forward if all arcs are followed in the forward direction
They are cycle is the origin is the same as the destination